

Activity: regression with multiple explanatory variables

Group members:

Data

In this class activity, we will try to predict a student's GPA in their first year of college, using their high school GPA and their SAT verbal score. The `FirstYearGPA` dataset, part of the `Stat2Data` package, includes the following variables on 219 students:

- **GPA**: First-year college GPA on a 0.0 to 4.0 scale
- **HSGPA**: High school GPA on a 0.0 to 4.0 scale
- **SATV**: Verbal/critical reading SAT score (between 200 and 800)

We are interested in the regression model

$$\text{GPA} = \beta_0 + \beta_1 \text{HSGPA} + \beta_2 \text{SATV} + \varepsilon$$

Here is the R output:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.6351217	0.2955033	2.149	0.03272	*
HSGPA	0.4975320	0.0750569	6.629	2.66e-10	***
SATV	0.0012283	0.0003373	3.641	0.00034	***

Questions

1. What is the estimated college GPA for a student with a high school GPA of 3.5, and a verbal SAT score of 600?

2. Is there any verbal SAT score for which we would predict a college GPA of at least 3.5, for a student with a high school GPA of 3.5?

3. Interpret the estimated slopes 0.498 and 0.0012.